SoC Design Report — Battery-Powered Wildlife Tracker

Build Your Own SoC: The Silicon Startup Challenge · EE250 · Internet of Things

Oct 12, 2026, 11:25 AM

TEAM

Leo

Design iterations evaluated: 1

85 / 100

Excellent

A well-balanced chip for the Battery-Powered Wildlife Tracker: every mandatory requirement is met, and the extra choices match this product's priorities.

DESIGN

Processor: ARM Cortex-M CPU (8 budget, 1 effort)

SRAM 256 KB (1b/0e) · QSPI external-flash interface (1b/2e) · Multiprotocol radio (BLE + 802.15.4) (3b/2e) · Integrated GNSS receiver (4b/2e) · 12-bit ADC (1b/0e) · SPI / I2C / UART interfaces (1b/0e) · Sleep controller (1b/1e)

Budget: 20 / 20 units

Effort: 8 / 10 units

SCORE BREAKDOWN

Required functionality	40 / 40
Product-specific suitability	13 / 20
Budget efficiency	9 / 15
Engineering effort	13 / 15
Architecture quality	10 / 10

MANDATORY REQUIREMENTS

- **MET** **Location tracking** — Integrated GNSS receiver; Serial I/O (UART) connects a…
- **MET** **Environmental & battery sensing** — ADC samples analog sensors and battery voltage
- **MET** **Wireless upload** — Multiprotocol radio (BLE)
- **MET** **Deep-sleep power management** — Sleep controller
- **MET** **Data logging** — SRAM buffer (lost if the battery runs out); External flash…

STRENGTHS

- An on-chip GNSS receiver removes a separate module and lets the chip manage positioning power directly.
- Uses a low-power radio (no Wi-Fi). LoRa reaches gateways kilometres away; BLE reaches phones and nearby gateways.
- Power domains with memory retention switch off everything except what is needed between readings.

TRADEOFFS

- Licensing used 8 of your 20 budget units, so premium IP options were hard to afford.
- Location data is protected only by software encryption: workable, but slower and more energy per byte.
- Without DSP extensions, data is processed in plain software or sent raw (more radio time).

SUGGESTED IMPROVEMENTS

- Swap Multiprotocol radio (BLE + 802.15.4) for Bluetooth LE and remove SRAM 256 KB: Uses a low-power radio (no Wi-Fi). LoRa reaches gateways kilometres away; BLE reaches phones and nearby gateways.
- Swap Integrated GNSS receiver for Crypto accelerator (AES/SHA): Hardware security protects location data — endangered-species locations attract poachers. The tradeoff: you give up Integrated positioning.

TEAM REFLECTION

In 2-4 sentences: why did your team choose this processor and these blocks, and what tradeoff did you accept?

(no reflection entered)